

SET-A

Name:	ID:	Section:
-------	-----	----------

Question 1 [C01] [10 Points]

The water park needs a program to manage visitor eligibility for various activities and special offers. Write a Java program that first takes the number of visitor sessions, followed by the number of visitors in each session. For each visitor, the program should further take the Activity Level (1 for Low, 2 for Moderate, 3 for High) and whether the visitor has visited the park previously (true for previous visitors, false for first-time visitors). The park offers the following activities and special offers: the **Lazy Pool** is available for visitors with an activity level of 2 or 3; the **Water Coaster** is available for visitors with an activity level of 3 who have visited the park before; and the **Special Offer** is available to first-time visitors. The program should count and display the total number of eligible visitors for each activity and for the Special Offer.

Sample Input	Sample Output
Enter number of visitor sessions: 2 Enter number of visitors in session 1: 3 Enter activity level of visitor 1: 2 Enter previous visit of visitor 1: true Enter activity level of visitor 2: 1 Enter previous visit of visitor 2: false Enter activity level of visitor 3: 2 Enter previous visit of visitor 3: false Enter number of visitors in session 2: 2 Enter activity level of visitor 1: 3 Enter previous visit of visitor 1: true Enter activity level of visitor 2: 2 Enter previous visit of visitor 2: false	Lazy Pool Riders: 4 Water Coaster Riders: 1 Visitors with Special Offer: 3 Explanation: For session 1, Visitor 1 qualifies for the Lazy Pool but not the Water Coaster due to an activity level of 2. Visitor 2 qualifies for the Special Offer as a first-time visitor. Visitor 3 qualifies for the Lazy Pool and Special Offer. For session 2, Visitor 1 qualifies for both the Lazy Pool and the Water Coaster. Visitor 2 qualifies for the Lazy Pool and the Special Offer.

SET-B

Name:	ID:	Section:
-------	-----	----------

Question 1 [C01] [10 Points]

You are asked to write a Java program for an amusement park that manages visitor eligibility for multiple rides and special offers. The program first takes the number of entry sessions from the management team. For each session, the program takes the number of visitors, followed by each visitor's height (in cm) and age (in years). The park has multiple rides with eligibility rules: the **Roller Coaster** requires a minimum height of 140 cm and a minimum age of 12 years, the **Ferris Wheel** allows visitors aged 5 years or older, and the **Bumper Cars** allow visitors aged 8 years or older with a maximum height of 200 cm for safety. Additionally, visitors under 12 years are eligible for a **Special Discount Offer**. The program should count the number of visitors who are eligible for each ride, as well as how many receive the discount. Finally, the program should display these counts after processing all sessions.

Sample Input	Sample Output
Enter number of sessions = 3 Enter number of visitors in session 1 = 3 Enter height of visitor 1: 150 Enter age of visitor 1: 14 Enter height of visitor 2: 120 Enter age of visitor 2: 10 Enter height of visitor 3: 160 Enter age of visitor 3: 20 Enter number of visitors in session 2 = 2 Enter height of visitor 1: 130 Enter age of visitor 1: 6 Enter height of visitor 2: 170 Enter age of visitor 2: 13 Enter number of visitors in session 3 = 1 Enter height of visitor 1: 180 Enter age of visitor 1: 9	Roller Coaster Riders: 3 Ferris Wheel Riders: 6 Bumper Cars Riders: 5 Visitors with Special Offer: 3 Explanation: Visitor 1 in session 1 qualifies for all rides, Visitor 2 qualifies for the Ferris Wheel and Bumper Cars and gets the Special Offer, and Visitor 3 qualifies for all rides. In session 2, Visitor 1 qualifies for the Ferris Wheel and the Special Offer, while Visitor 2 qualifies for all rides. In session 3, the visitor qualifies for the Ferris Wheel and Bumper Cars and receives the Special Offer.

Solution & Rubric

In case of any circumstance not present in the rubrics, feel free to discuss in the assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

SET - [A]

SOLUTION

```
import java.util.Scanner;

public class WaterPark {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int lazyPool = 0, waterCoaster = 0, specialOffer = 0;

        System.out.print("Enter number of visitor sessions: ");
        int sessions = sc.nextInt();

        for (int i = 1; i <= sessions; i++) {
            System.out.print("Enter number of visitors in session " + i + ": ");
            int visitors = sc.nextInt();

            for (int j = 1; j <= visitors; j++) {
                System.out.print("Enter activity level of visitor " + j + ": ");
                int activityLevel = sc.nextInt();
                System.out.print("Enter previous visit of visitor " + j + ": ");
                boolean previousVisit = sc.nextBoolean();

                if (activityLevel == 2 || activityLevel == 3) {
                    lazyPool++;
                }

                if (activityLevel == 3 && previousVisit==true) {
                    waterCoaster++;
                }

                if (previousVisit==false) {
                    specialOffer++;
                }
            }
        }

        System.out.println("Lazy Pool Riders: " + lazyPool);
        System.out.println("Water Coaster Riders: " + waterCoaster);
        System.out.println("Visitors with Special Offer: " + specialOffer);
    }
}
```

RUBRIC

SL	Points to Meet	Marks [10]
1	The program takes input for sessions, number of visitors, activity level, and previous visit	1
2	Correctly calculates eligibility for all activities (Lazy Pool, Water Coaster)	3
3	Correctly calculates eligibility for Special Offer	1
4	Displays total counts for all activities and Special Offers correctly	2
5	Proper use of nested loops and conditionals	3

SET - [B]**SOLUTION**

```
import java.util.Scanner;

public class AmusementPark {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int rollerCoaster = 0, ferrisWheel = 0, bumperCars = 0, specialOffer = 0;

        System.out.print("Enter number of sessions = ");
        int sessions = sc.nextInt();

        for (int i = 1; i <= sessions; i++) {
            System.out.print("Enter number of visitors in session " + i + "= ");
            int visitors = sc.nextInt();

            for (int j = 1; j <= visitors; j++) {
                System.out.print("Enter height of visitor " + j + ": ");
                int height = sc.nextInt();
                System.out.print("Enter age of visitor " + j + ": ");
                int age = sc.nextInt();

                if (height >= 140 && age >= 12) {
                    rollerCoaster++;
                }

                if (age >= 5) {
                    ferrisWheel++;
                }

                if (age >= 8 && height <= 200) {
                    bumperCars++;
                }

                if (age < 12) {
                    specialOffer++;
                }
            }
        }

        System.out.println("Roller Coaster Riders: " + rollerCoaster);
        System.out.println("Ferris Wheel Riders: " + ferrisWheel);
        System.out.println("Bumper Cars Riders: " + bumperCars);
        System.out.println("Visitors with Special Offer: " + specialOffer);
    }
}
```

RUBRIC		
SL	Points to Meet	Marks [10]
1	The program takes input for sessions, number of visitors, height, and age	1
2	Correctly calculates eligibility for all rides (Roller Coaster, Ferris Wheel, Bumper Cars)	3
3	Correctly calculates Special Offer eligibility	1
4	Displays total counts for all rides and Special Offers correctly	2
5	Proper use of nested loops	3